

UPS Delivery Crash Risk Estimator

This app estimates crash severity risk using the final Random Forest workflow from the modeling notebook. Inputs were simplified into driver-facing categories for weather, light, road surface, trafficway type, and hour of day. Important limitation: this model reflects the uploaded SMS crash dataset and has low overall predictive accuracy, despite stronger recall for fatal cases.

Weather

rain

Light Condition

night

Road Surface

wet

Trafficway Type

undivided

Hour of Day

22

0

23

Clear

Submit

Primary Predicted Severity

Fatal

Risk Summary and Guidance

Primary predicted outcome: Fatal

Driver guidance: High-alert scenario. Reduce speed, extend following distance, and avoid risky maneuvers.

Predicted Probability Breakdown

Severity	Probability (%)
Fatal	64.81
Property_Damage	20.03
Injury	15.17

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Weather

rain

Light Condition

day

Road Surface

wet

Trafficway Type

divided

Hour of Day

01423

Clear

Submit

Primary Predicted Severity

Injury

Risk Summary and Guidance

Primary predicted outcome: Injury

Driver guidance: Elevated injury risk. Stay alert, avoid aggressive lane changes, and be cautious at conflict points.

Predicted Probability Breakdown

Severity	Probability (%)
Injury	57.6
Property_Damage	42.4
Fatal	0

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Weather

wind

Light Condition

day

Road Surface

dry

Trafficway Type

divided

Hour of Day

01123

Clear

Submit

Primary Predicted Severity

Property_Damage

Risk Summary and Guidance

Primary predicted outcome: Property_Damage

Driver guidance: Lower severity than the other categories, but caution is still needed.

Predicted Probability Breakdown

Severity	Probability (%)
Property_Damage	89.55
Injury	6.56
Fatal	3.89

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